BLOCKCHAIN INVESTIGATION · FLOW-OF-FUNDS REPORT

Following the Bybit heist: three laundering branches, traced hop by hop

The $1.4–1.5 billion theft from Bybit on 21 February 2025 is the largest crypto heist on record. This report follows three representative branches of the stolen ETH from the exploiter's wallet to their exits, checking each hop against the raw transaction record.

PREPARED BY	REPORT DATE	INCIDENT DATE	ATTRIBUTION	CHAINS
Treylon Wofford	7 November 2026	21 Feb 2025, 14:16 UTC	DPRK "TraderTraitor" (FBI)	Ethereum → Bitcoin
HOPS VERIFIED				
20 transactions				

Bottom line

- Within **14 minutes** of the theft, the exploiter funded a second wallet. Within **27 minutes**, that wallet was turning stolen Lido stETH into ETH through DEX aggregators. In 71 minutes it received **98,047.84 ETH** in swap proceeds, which roughly matches the ~98,375 stETH and mETH stolen.
- Branch A moved **257.27 ETH** from the exploiter to THORChain in **16 minutes** through two single-use wallets. The THORChain deposit memo shows a swap to a Bitcoin address.
- Branch C split funds from a labeled exploiter wallet, then **merged 2,083 ETH from six wallets** into one address. It sent a 1 ETH test, moved the rest to a holding wallet, and left it untouched for **70 days** before moving it again in May 2025.
- Third parties ran **address poisoning** against the launderers. About 98% of the exploiter's 4,418 transactions are incoming dust from look-alike addresses. That noise can mislead analysts who match addresses by their first and last characters.
- I assess with **high confidence** that the funds in all three branches came from the Bybit theft. Each hop below links to a transaction anyone can check.

13m 36s
Theft to gas funding of Exploiter 2

27m 12s
Theft to first swap proceeds (ParaSwap)

16m 12s
Exploiter 1 to THORChain, branch A

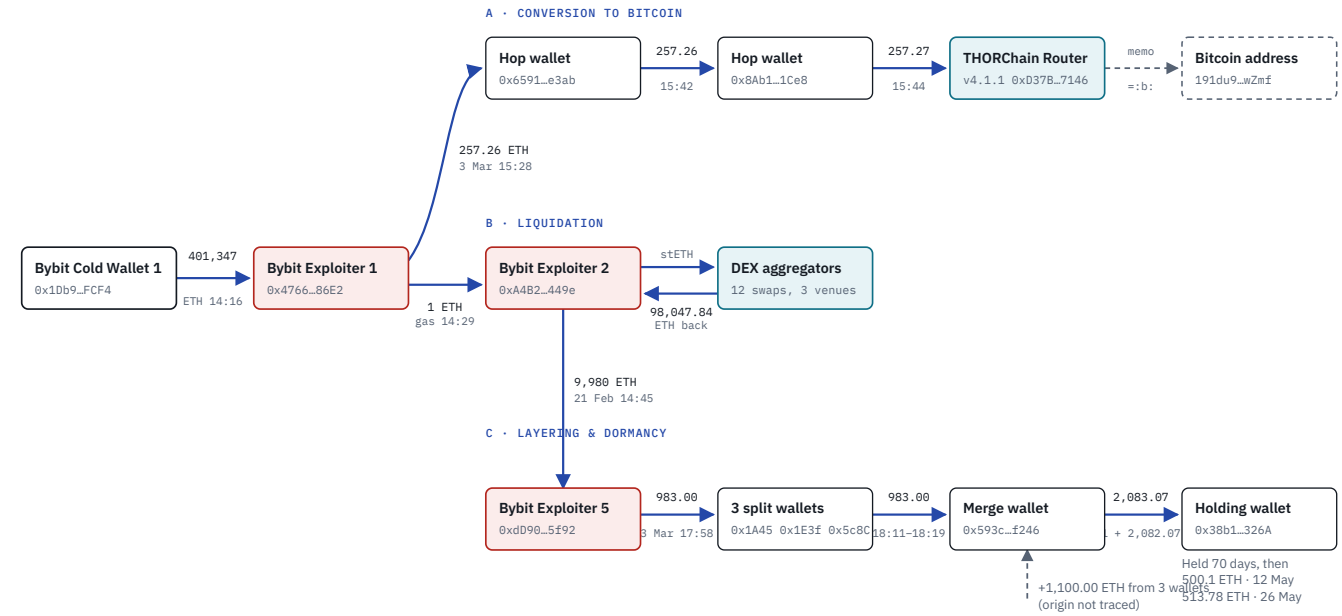
69d 23h
Dormancy of merged 2,082 ETH, branch C

1 Background

Bybit's Ethereum cold wallet (`0x1Db92e2EeBC8E0c075a02BeA49a2935BcD2dFCF4`) was a Safe multisig. Attackers compromised a Safe{Wallet} developer environment and injected JavaScript into the Safe web app that activated only for Bybit's wallet. When signers approved what looked like a routine transfer at 14:13:35 UTC, they signed a contract upgrade that handed control to the attacker.

Ninety-six seconds later, a `sweepETH` call moved **401,346.77 ETH** to `0x47666Fab8bd0Ac7003bce3f5C3585383F09486E2`, labeled "Bybit Exploiter 1" on Etherscan. Public reporting puts the total haul at about 401,000 ETH, 90,375 stETH, 15,000 cmETH and 8,000 mETH. mETH Protocol later recovered the 15,000 cmETH. On 26 February 2025 the FBI attributed the theft to North Korea ("TraderTraitor") and published 51 Ethereum addresses for the industry to block.

2 Flow of funds



Amounts in ETH; times in UTC. Branch C amounts are combined across the three split wallets. Full addresses and transaction links are in the evidence log.

3 Scope, method and confidence

The theft fanned out to thousands of addresses, so tracing every unit is a job for a team with commercial tooling. I chose three branches, one for each phase of the laundering: **liquidation** (turning staked ETH into plain ETH), **conversion** (moving ETH off Ethereum) and **layering** (splitting and merging to break the trail).

I checked every hop against the transaction record on Etherscan: sender, receiver, amount, block and timestamp. I linked hops through exact amount matches, timing and Etherscan's "funded by" field. I decoded the THORChain deposit memo from the transaction's input data.

Figures from news and vendor reports are marked *Reported* and kept apart from my own findings.

Confirmed
A direct transfer I viewed in the transaction record, with matching addresses and amount.

Reported
Stated in public reporting (FBI, Verichains, analytics firms). Not independently re-verified here.

Not traced
Named in the data, but this investigation did not follow it further.

4 Evidence log

#	TIME (UTC)	FROM	TO	AMOUNT	TRANSACTION	STATUS
THEFT · 21 FEB 2025						
1	14:13:35	Safe signers	Cold Wallet 1 proxy: malicious implementation upgrade	n/a	0x46deef0f…7882	Reported
2	14:16:11	Bybit Cold Wallet 1	Bybit Exploiter 1 (sweepETH)	401,346.77 ETH	0xb61413c4…072c	Confirmed
BRANCH B · LIQUIDATION · 21 FEB 2025						
3	14:29:47	Bybit Exploiter 1	Bybit Exploiter 2 (gas)	1.00 ETH	0xdd5cd734…e4bc	Confirmed
4	14:42:47	Bybit Exploiter 2	Approves ParaSwap Delta V2 to spend stETH	10,000 stETH	0x8ece559e…e728	Confirmed
5	14:43:23–15:27:47	ParaSwap (8), Uniswap (3), DODO (1)	Bybit Exploiter 2: swap proceeds	98,047.84 ETH	12 internal txs	Confirmed
6	14:45:11	Bybit Exploiter 2	Bybit Exploiter 5	9,980.00 ETH	0xf7900c28…e0c9	Confirmed
BRANCH A · CONVERSION TO BITCOIN · 3 MAR 2025						
7	15:28:35	Bybit Exploiter 1	0x6591c65321E3c5749A2A0ecb7E9F1aEb4FA3e3ab	257.26 ETH	0xf4a3ab13…cd92	Confirmed
8	15:42:59	0x6591…e3ab	0x8Ab1D1d3E7db399835172576CcE0bD1C200a1Ce8	257.26 ETH	0x3463e5c0…8c3c	Confirmed
9	15:44:47	0x8Ab1…1Ce8	THORChain Router v4.1.1 (depositWithExpiry)	257.27 ETH	0x16ed29f9…fa1a	Confirmed
10	after 15:44	THORChain	BTC 191du9ivk3k5YekuXFgtkoVei6pK76wZmf (memo destination)	n/a	Bitcoin outbound	Not traced
BRANCH C · LAYERING AND DORMANCY · 3 MAR TO 26 MAY 2025						
11	17:58:59	Bybit Exploiter 5	0x1A4559d21215cE38e8eDe6aec310316a84c5702f	561.06 ETH	0x167228e4…9287	Confirmed
12	17:59:11	Bybit Exploiter 5	0x1E3f59f0586e576659F1F42f69590D3d5ABE8f25	286.29 ETH	0x3b6504cb…2d28	Confirmed
13	18:13:47	Bybit Exploiter 5	0x5c8CD411A4a85145Dd6Bb8a049C16f0e537aCffa	135.65 ETH	0xa2e4935c…f9cf	Confirmed
14	18:11:59	0x1A45…702f	Merge wallet 0x593c22d3Bc59EB8A3CB7317F31fc9224b857f246	561.06 ETH	0x7cfdaf4a…64d4	Confirmed
15	18:16:11	0x1E3f…8f25	Merge wallet	286.29 ETH	0xe73afadc…ebb0	Confirmed

#	TIME (UTC)	FROM	TO	AMOUNT	TRANSACTION	STATUS
16	18:19:23	0x5c8C…cffa	Merge wallet	135.65 ETH	0xc20f739a…7ccf	Confirmed
17	17:58:47–18:42:59	3 other wallets (0xD710..., 0x1C6C..., 0x5445...)	Merge wallet	1,100.00 ETH	3 transfers	Not traced
18	18:57:47	Merge wallet	Holding wallet 0x38b1190177e8178452b851064dF73341E94A326A (test)	1.00 ETH	0x03be6ae4…6114	Confirmed
19	19:58:47	Merge wallet	Holding wallet	2,082.07 ETH	0x93ec8588…b95a	Confirmed
20	12 May 19:54:59	Holding wallet	0xDdb8993F0EF7629aAb5362DaCef2C2aAc3b7c5A9	500.10 ETH	0xcf5ae2cf…49a5	Confirmed
21	26 May 14:09:35	Holding wallet	0x8aFF2582A93343f1896E70abAe90662BE189ec68	513.78 ETH	0x6a40892a…2738	Confirmed

Row 1 comes from Verichains' technical analysis. Row 5 adds up the 12 internal transfers to Exploiter 2 shown on Etherscan. Etherscan shows only the latest 25 transactions per address, so other activity by these wallets may exist.

5 Findings

01 Liquid staking tokens were dumped first, through aggregators

Exploiter 2 got 1 ETH of gas 13 minutes after the theft, approved ParaSwap's Delta V2 contract to spend stETH, and collected 98,047.84 ETH in 12 swaps over 44 minutes. That total is within 0.4% of the 98,375 stETH and mETH reported stolen. Staked ETH can be frozen or blocklisted by its issuer, so it was the riskiest asset to hold. Converting it first and fast follows from that. *Assessment: high confidence these swaps liquidated the stolen stETH and mETH. I did not view the token transfer that moved stETH into Exploiter 2, so that link rests on the approval, the amounts and the timing.*

02 Single-use wallets fed THORChain within minutes

In branch A, each of two fresh wallets received exactly 257.26 ETH and passed it on within 14 minutes and 2 minutes. Each kept only dust. The final deposit's memo, `=:b:19 1du9ivk3k5YekuXFgtkoVei6pK76wZmf:0/1/0:dx:10`, asks THORChain to swap the ETH to native BTC, deliver it to that address and run it as a streaming swap. The memo also carries an affiliate tag. Public reporting attributes about 72% of the converted ETH to THORChain, and branch A shows that route at the level of a single transaction.

03 Split, merge, test, then wait

Exploiter 5 sent ETH to three wallets that each forwarded the exact amount to one merge wallet within 20 minutes. Three more wallets, not traced here, added 1,100 ETH, for 2,083.00 ETH in total. The merge wallet sent a 1 ETH test to a new address, then the remaining 2,082.07 ETH an hour later. That address then sat untouched for 69 days and 23 hours. It reappeared in May 2025 with transfers of 500.1 ETH and 513.78 ETH. *Assessment: the dormancy points to the launderers waiting for attention to fade. 983.00 ETH (47%) of the merged funds trace directly to Exploiter 5. The rest arrived at the same time and in the same pattern but is unverified.*

04 The launderers were themselves being poisoned

Exploiter 1 has 4,418 transactions, but its nonce shows it sent only about 62. Roughly 4,350 (about 98%) are incoming dust or zero-value transfers. Many come from addresses that copy the first and last characters of wallets the launderers had just used, hoping someone copies the wrong address from their history. The same pattern appears around the holding wallet in May. For investigators, this noise inflates transaction counts and can mislead any tool or analyst that matches addresses by prefix and suffix. Several zero-value transfers also appear to come from an address labeled "FixedFloat Dep". Zero-value token transfers can be spoofed, so I treat those as noise, not as evidence of an exchange deposit.

USED BY LAUNDERER	REAL COUNTERPARTY	LOOK-ALIKE THAT SENT DUST
Exploiter 1, hop 7	`0x6591`c65321E3c5749A2A0ecb7E9F1aEb4FA3`e3ab`	`0x6591`d87DE235e5EE78688Dd3858c2f4f38f7`e3ab`
Hop wallet 0x6591, hop 8	`0x8Ab1`D1d3E7db399835172576CcE0bD1C200a`1Ce8`	`0x8aB1`Ceded9796902C9554359ccd396F5d2Ec`1ce8`
Exploiter 1, 3 Mar	`0x9Bb1`4c34740996c8Ce9824bc8E09161972674`E33`	`0x9Bb1`1917b3aD3C72D9e5eB5C5FFE0a914A0c4`E33`
Exploiter 1, 3 Mar	`0xb490`dAB6E0b7Df4DbF3b1b607E0E940BC7FA`31C9`	`0xb490`c2362904fc11B7975ba681387daA0eB0`31C9`
Holding wallet, hop 20	`0xDdb8`993F0EF7629aAb5362DaCef2C2aAc3b7`c5A9`	`0xDdB8`a33659FF7c973Cc8b43744909355bcda`c5A9`

6 The wider picture Reported

- About 401,000 ETH plus ~113,000 in staked ETH derivatives were stolen, worth $1.4–1.5 billion at the time.
- On the first day, the haul was split into roughly 40–50 wallets. Within a week the number of addresses reached the thousands, and the FBI's notice lists 51 key Ethereum addresses.
- Arkham and others report that about 72% of the converted ETH crossed to Bitcoin through THORChain. eXch, a no-KYC swap service, handled about 79,655 ETH (about 16%).
- On 4 March 2025, Bybit's CEO said 77% of the funds were still traceable, 20% had "gone dark" and 3% had been frozen. Tether and Circle froze stablecoins linked to the hack, and Chainflip refused deposits from flagged addresses.

7 Open leads

1. Look up THORChain's outbound transaction for inbound hash `0x16ed29f9…fa1a`, then follow BTC address `191du9ivk3k5YekuXFgtkoVei6pK76wZmf` on Bitcoin. Watch for consolidation, mixing or exchange deposits.
2. Follow the May 2025 outflows from the holding wallet (500.1 ETH and 513.78 ETH) and the rest of its 138 transactions.
3. Trace back the three wallets that put 1,100 ETH into the merge wallet, to confirm they come from the same theft.
4. Pull the stETH and mETH token transfer history of Exploiter 2 to close the one inferred link in branch B.
5. Screen every address in this report against the FBI's list and commercial attribution data.

8 Recommendations for compliance teams

- **Screen for indirect exposure as well as direct.** Every wallet in branches A and C is unlabeled, one to three hops from a listed exploiter address. Screening only for direct contact misses them.
- **Treat dormant wallets as live risk.** Branch C funds stayed still for 70 days. Alert on sudden activity from wallets that are a few hops from a known hack.
- **Hold BTC deposits from cross-chain swaps for review** when they arrive in the weeks after a major hack, especially from THORChain streaming swaps.
- **Compare full addresses, never just the prefix and suffix,** in investigations, withdrawal allowlists and case notes. Address poisoning is aimed at exactly that shortcut.

9 Addresses for screening

ADDRESS	CHAIN	ROLE
0x1Db92e2EeBC8E0c075a02BeA49a2935BcD2dFCF4	Ethereum	Victim: Bybit Cold Wallet 1
0x0fa09C3A328792253f8dee7116848723b72a6d2e	Ethereum	Attacker: sent the sweepETH call
0x47666Fab8bd0Ac7003bce3f5C3585383F09486E2	Ethereum	Bybit Exploiter 1: received 401,346.77 ETH
0xA4B2Fd68593B6F34E51cB9eDB66E71c1B4Ab449e	Ethereum	Bybit Exploiter 2: stETH liquidation
0xdD90071D52F20e85c89802e5Dc1eC0A7B6475f92	Ethereum	Bybit Exploiter 5: branch C source
0x6591c65321E3c5749A2A0ecb7E9F1aEb4FA3e3ab	Ethereum	Branch A hop wallet
0x8Ab1D1d3E7db399835172576CcE0bD1C200a1Ce8	Ethereum	Branch A: deposited to THORChain
191du9ivk3k5YekuXFgtkoVei6pK76wZmf	Bitcoin	Branch A: THORChain memo destination
0x1A4559d21215cE38e8eDe6aec310316a84c5702f	Ethereum	Branch C split wallet
0x1E3f59f0586e576659F1F42f69590D3d5ABE8f25	Ethereum	Branch C split wallet
0x5c8CD411A4a85145Dd6Bb8a049C16f0e537aCffa	Ethereum	Branch C split wallet
0x593c22d3Bc59EB8A3CB7317F31fc9224b857f246	Ethereum	Branch C merge wallet (2,083 ETH)
0x38b1190177e8178452b851064dF73341E94A326A	Ethereum	Branch C holding wallet (dormant 70 days)
0xDdb8993F0EF7629aAb5362DaCef2C2aAc3b7c5A9	Ethereum	Branch C: received 500.1 ETH, 12 May 2025
0x8aFF2582A93343f1896E70abAe90662BE189ec68	Ethereum	Branch C: received 513.78 ETH, 26 May 2025

10 How this report was made

AI helped with this investigation: gathering public reporting, reading block explorer records, decoding the THORChain memo from hex and drafting the text. The investigative decisions are in the report and can be checked: which branches to follow, what counts as a confirmed link, where the evidence stops. Every confirmed hop links to its transaction so a reader can verify it independently. Data was collected on 7 November 2026 from public sources only.

11 Sources

1. FBI, PSA I-022625-PSA: North Korea responsible for $1.5 billion Bybit hack (26 Feb 2025)
2. Verichains, Technical analysis of the Bybit wallet exploit
3. Nansen, Bybit hack: what happened and where the funds are going
4. Arkham, Lazarus bridged $1.3B of Bybit ETH to Bitcoin (5 Mar 2025)
5. AMLCrypto, Event chronology of the Bybit hack
6. The Block, Bybit CEO says 20% of $1.4 billion theft has "gone dark"
7. Etherscan transaction and address records, linked in the evidence log

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TLP:CLEAR. Built entirely from public blockchain data and published reporting. Addresses described as hop, merge or holding wallets are identified by their on-chain behavior, not by the identity of their owners.

Prepared by Treylon Wofford · 7 November 2026